A

light aluminum oxide is also included. The PhEur 6.3 includes a specification for hydrated aluminum oxide that contains the equivalent of 47.0–60.0% of Al_2O_3.

The EINECS number for aluminum oxide is 215-691-6.

Table I: JPE specification for aluminum oxide.[6]

Test	JPE 2004
Identification	+
Water-soluble substances	+
Heavy metals	$\leqslant$30 ppm
Lead	$\leqslant$30 ppm
Arsenic	$\leqslant$5 ppm
Loss on drying	$\leqslant$1.5%
Loss on ignition	$\leqslant$2.5%
Assay	$\geqslant$96.0%

19 Specific References

1 Rupprecht H. Processing of potent substances with inorganic supports by imbedding and coating. *Acta Pharm Technol* 1980; **26**: 13–27.
2 Lewis RJ, ed. *Sax's dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 136.
3 National Poisons Information Service 1997. Aluminium oxide. http://www.intox.org/databank/documents/chemical/alumoxde/ukpid33.htm (accessed 16 January 2009)
4 Health and Safety Executive. *EH40/2005: Workplace Exposure Limits*. Sudbury: HSE Books, 2005 (updated 2007). http://www.hse.gov.uk/coshh/table1.pdf (accessed 5 February 2009).
5 JT Baker (2007). Material safety data sheet: Aluminium oxide. http://www.jtbaker.com/msds/englishhtml/a2844.htm (accessed 5 February 2009).
6 Japan Pharmaceutical Excipients Council. *Japanese Pharmaceutical Excipients 2004*. Tokyo: Yakuji Nippo, 2004; 67–68.

20 General References

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21 Author

T Farrell.

22 Date of Revision

5 February 2009.

Aluminum Phosphate Adjuvant

1 Nonproprietary Names

None adopted.

2 Synonyms

Adju-Phos; aluminum hydroxyphosphate; aluminium hydroxyphosphate; *Rehydraphos*.

3 Chemical Name and CAS Registry Number

Aluminum phosphate [7784-30-7]

4 Empirical Formula and Molecular Weight

$Al(OH)_x(PO_4)_y$

The molecular weight is dependent on the degree of substitution of phosphate groups for hydroxyl groups.

5 Structural Formula

Aluminum phosphate adjuvant occurs as a precipitate of amorphous aluminum hydroxide in which some sites contain phosphate groups instead of hydroxyl. Both hydroxyl and phosphate groups are exposed at the surface. The hydroxyl groups produce a pH-dependent surface charge by accepting a proton to produce a positive site, or donating a proton to produce a negative site. The pH-dependent surface charge is characterized by the point of zero charge, which is equivalent to the isoelectric point in protein chemistry. The surface hydroxyl groups may also undergo ligand exchange with fluoride, phosphate, carbonate, sulfate, or borate anions.

Aluminum phosphate adjuvant is not a stoichiometric compound. Rather, the degree of phosphate group substitution for hydroxyl groups depends on the precipitation recipe and conditions.

6 Functional Category

Adsorbent; vaccine adjuvant.

7 Applications in Pharmaceutical Formulation or Technology

Aluminum phosphate adjuvant is used in parenteral human and veterinary vaccines.[1] It activates Th2 immune responses, including IgG and IgE antibody responses.

8 Description

Aluminum phosphate adjuvant is a white hydrogel that sediments slowly and forms a clear supernatant.

9 Pharmacopeial Specifications

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10 Typical Properties

Acidity/alkalinity pH = 6.0–8.0
Al : P atomic ratio 1.0–1.4 : 1.0
Aluminum (%) 0.5–0.75
Particle size distribution Primary particles are platy with an average diameter of 50 nm. The primary particles form aggregates of 1–10 μm.
Point of zero charge pH = 4.6–5.6, depending on the Al : P atomic ratio.
Protein binding capacity >0.6 mg lysozyme/mg equivalent Al_2O_3
Solubility Soluble in mineral acids and alkali hydroxides.
X-ray diffractogram Amorphous to x-rays.

11 Stability and Storage Conditions

Aluminum phosphate adjuvant is stable for at least 2 years when stored at 4–30°C in well-sealed inert containers. It must not be allowed to freeze as the hydrated colloid structure will be irreversibly damaged.

12 Incompatibilities

The point of zero charge is related directly to the Al : P atomic ratio. Therefore, the substitution of additional phosphate groups for hydroxyl groups will lower the point of zero charge. Substitution of carbonate, sulfate, or borate ions for hydroxyl groups will also lower the point of zero charge.

13 Method of Manufacture

Aluminum phosphate adjuvant is formed by the reaction of a solution of aluminum chloride and phosphoric acid with alkali hydroxide.

14 Safety

Aluminum phosphate adjuvant is intended for use in parenteral vaccines and is generally regarded as safe. It may cause mild irritation, dryness, and dermatitis on skin contact. It may also cause redness, conjunctivitis, and short-term mild irritation on eye contact. Ingestion of large amounts of aluminum phosphate adjuvant may cause respiratory irritation with nausea, vomiting, and constipation. Inhalation is unlikely, although the dried product may cause respiratory irritation and cough. Type I hypersensitivity reactions following parenteral administration have also been reported.(2)

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Eye protection and gloves are recommended.

16 Regulatory Status

GRAS listed. Accepted for use in human and veterinary vaccines in Europe and the USA. The limits for use in human vaccines are 0.85 mg aluminum/dose (FDA) and 1.25 mg aluminum/dose (WHO). There are no established limits for use in veterinary vaccines. Reported in the EPA TSCA Inventory.

17 Related Substances

Aluminum hydroxide adjuvant.

18 Comments

The USP 32 monograph for aluminum phosphate ($AlPO_4$) gel describes aluminum phosphate, which is used as an antacid, not as a vaccine adjuvant.

19 Specific References

1 Shirodkar S *et al*. Aluminum compounds used as adjuvants in vaccines. *Pharm Res* 1990; 7: 1282–1288.
2 Goldenthal KL *et al*. Safety evaluation of vaccine adjuvants. *AIDS Res Hum Retroviruses* 1993; **9**: S47–S51.

20 General References

Hem SL, Hogenesch H. Aluminum-containing adjuvants: properties, formulation, and use. Singh M, ed. *Vaccine Adjuvants and Delivery Systems*. New York: Wiley, 2007; 81–114.

Gupta RK *et al*. Adjuvant properties of aluminum and calcium compounds. Powell MF, Newman MJ, eds. *Vaccine Design*. New York: Plenum, 1995; 229–248.

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21 Authors

SL Hem, PB Klepak, EB Lindblad.

22 Date of Revision

19 February 2009.

Ammonia Solution

1 Nonproprietary Names

BP: Strong Ammonia Solution

PhEur: Ammonia Solution, Concentrated

USP-NF: Strong Ammonia Solution

2 Synonyms

Ammoniaca; ammoniacum; ammoniae solution concentrata; aqua ammonia; concentrated ammonia solution; spirit of hartshorn; stronger ammonia water.

3 Chemical Name and CAS Registry Number

Ammonia [7664-41-7]

4 Empirical Formula and Molecular Weight

NH_3 17.03

5 Structural Formula

See Section 4.

6 Functional Category

Alkalizing agent.

7 Applications in Pharmaceutical Formulation or Technology

Ammonia solution is typically not used undiluted in pharmaceutical applications. Generally, it is used as a buffering agent or to adjust the pH of solutions. Most commonly, ammonia solution (the